

3(a)	<i>max 2 if only one section attempted</i> <i>three from:</i> <i>multilayered epidermis</i> reduces, cuticular transpiration / loss of water <u>vapour</u> through cuticle ; increases distance for diffusion (of water vapour to cuticle) ; ref. to protection from heat from sunlight and reduced evaporation (from spongy mesophyll cells) ; <i>stomatal crypts</i> creates area of moist air / AW ; I traps water minimises effect of, external air currents / wind ; reduces / less steep, water potential gradient (between intercellular air space and external environment) ; A water vapour potential gradient A diffusion gradient if referenced to water vapour ref. to only on lower surface / shaded, and reduced evaporation (from spongy mesophyll cells) ; A lower temperature reduces rate of diffusion of water vapour (out via stomata)	3
3(b)	phloem sieve tube element(s) ; A phloem sieve tube (cells) A sieve elements I companion / transfer, cells	1

Question	Answer	Marks
3(c)	<i>allow named assimilates</i> <i>source</i> <i>in context of assimilates</i> site of synthesis ; A described e.g. photosynthesis in leaf / storage compound breakdown in roots <i>sink</i> <i>in context of, via phloem / from source / from leaf</i> I nutrients / substances, unqualified <i>if assimilates / named assimilate, not stated when explaining source</i> site where assimilates are stored A described e.g. roots for storage of starch or area where, growth occurs / assimilates are used for growth A described e.g. growth in developing buds / growth of immature leaf or area that receives, assimilates / AW ; A area where sucrose unloaded I place that needs assimilates I place where assimilates are used, unless qualified	2
3(d)(i)	change in the, sequence / AW, of, bases / nucleotides (in, the DNA / the gene / <i>cyFBP</i>) ;	1

Question	Answer	Marks
3(d)(ii)	<i>one valid suggestion that prevents polypeptide synthesis or prevents synthesis of functional enzyme:</i> <i>e.g.</i> no mRNA produced ; A transcription, does not / cannot, occur mRNA produced unable to attach to ribosome ; A translation, does not / cannot, occur no start codon on mRNA ; shortened polypeptide chain produced ; A ref. to STOP codon premature chain termination mutation / nonsense mutation ; polypeptide produced cannot fold to form, tertiary structure / active site ; R changes shape of active site preventing function polypeptide produced cannot form quaternary structure of protein ;	1
3(e)(i)	<i>accept cell for lymphocyte throughout</i> <i>four from:</i> 1 (primary) immune response ; 2 cyFBPase / enzyme, is antigen ; 3 formation of, APC / antigen-presenting cell ; A antigens presented on surface of macrophages 4 antigen, recognition / binding (in context of B-, or T-lymphocytes) ; 5 detail ; e.g. clonal selection have receptors, complementary / specific, to, antigen / cyFBPase A immunoglobulin / antibody, as receptors for B-lymphocytes A surface molecules as receptors for T-lymphocytes 6 divide by mitosis / clonal expansion ; 7 (specific) B-lymphocytes form plasma cells ; 8 plasma cells, synthesise / secrete, specific antibody / antibody to cyFBPase ; 9 T-helper / Th, lymphocyte, secretes, cytokine / interleukin ; 10 stimulates, humoral / B-lymphocyte, response ;	4
3(e)(ii)	no binding of (monoclonal) antibody to, antigen / cyFBPase / enzyme / AW ;	1

Question	Answer	Marks
3(f)	<i>two from:</i> (less cyFBPase / enzyme, so) less sucrose synthesised ; A no enzyme to synthesise sucrose less sucrose transported (source to sink) ; A less sucrose unloaded at sink ; ref. to less glucose for synthesis of starch / AW (on context of sucrose hydrolysis) ;	2

Question	Answer	Marks
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